SCIENTIFIC COMMITTEE
SEVENTEENTH REGULAR SESSION

Electronic Meeting
11-19 August 2021

Tuna fisheries bycatch and climate change in the western tropical Pacific Ocean

WCPFC-SC17-2021/EB-IP-11
5 July 2021

Allain, V.[1], Griffiths, S.[2], MacDonald, J.[1], Wabnitz, C.[3,4], Pilling, G.M. [1], Nicol, S. [1]

[1] Pacific Community (SPC)
[2] Inter-American Tropical Tuna Commission (IATTC)
[3] Stanford University
[4] University of British Columbia (UBC)

Tuna fisheries bycatch and climate change in the western tropical Pacific Ocean

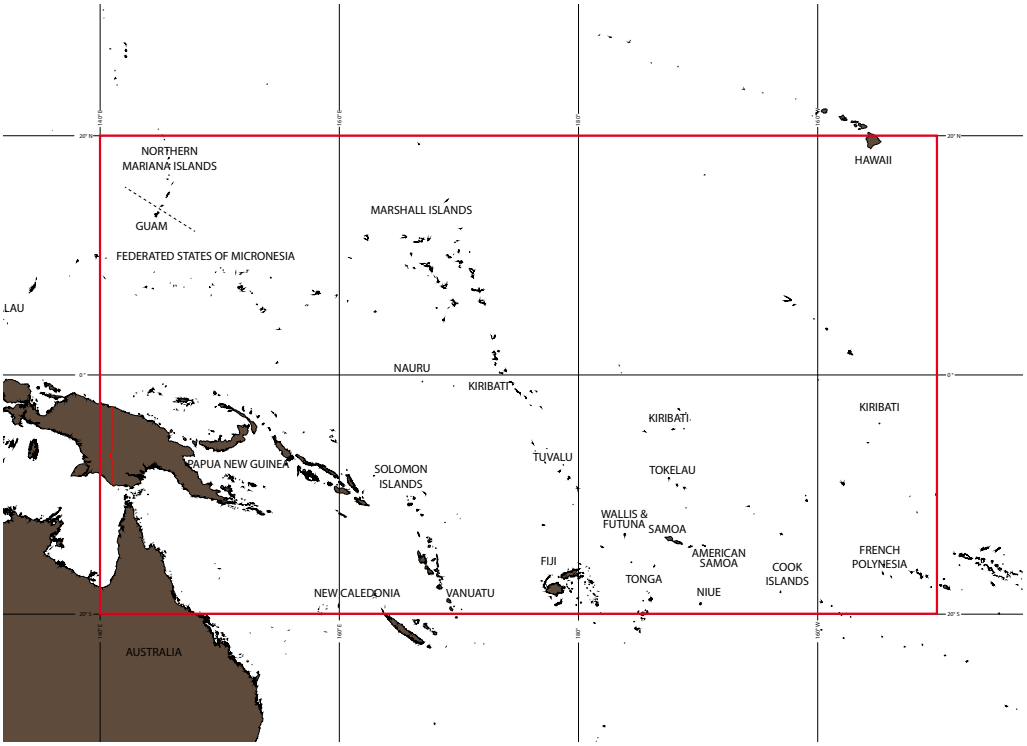
Project purpose: to inform Pacific Island countries and territories of the potential effects of climate change and tuna fishing on tropical western and central Pacific Ocean edible bycatch species.

Context

Employing an ecosystem approach to fisheries management that considers climate change is a crucial step towards the sustainable management of tuna resources and their wider ecosystem in the western and central Pacific Ocean. Purpose-built ecosystem models can provide a holistic view of the system and be used to explore the potential impacts of climate change and fishing on target and other species. This document summarizes the results of the application of such an approach to the tropical western and central Pacific Ocean with a focus on edible bycatch species caught within tuna fisheries (marlins, swordfish, wahoo, mahi mahi, rainbow runner, barracuda, and opah). These findings may inform regional decision-making around ecological sustainability and food security in the region.

Key messages

- Climate change has a negative impact on edible bycatch species biomass with the strongest effects predicted after 2050;
- Climate change and fishing effects interact. The ecosystem food web is sensitive to the synergistic effect of both factors, and for most species, climate change exacerbates the negative effect of any increase in fishing mortality;
- Long-lived species are particularly sensitive to climate change and fishing mortality;
- Ecosystem models are useful tools to support sustainable fisheries management as they allow the evaluation of the effects of interactions by multiple drivers. In many cases, these effects are indirect and modified by food web connections that are not represented in single species models.



Ecosystem model of the tropical western and central Pacific Ocean (140°E–150°W and 20°N–20°S)

Recommendations

- Ongoing and representative fisheries monitoring programmes that include data collection for bycatch species are key to support ecosystem approaches to fisheries management.
- Fisheries management measures are key tools for mitigating and adapting to climate change impacts and supporting the sustainable use of marine resources. However, the synergistic effects of climate change, and changes in fishing mortality through regulation of different fisheries can lead to unexpected outcomes for edible bycatch species. Careful consideration will be required when developing management measures to ensure they achieve objectives for the target species, edible bycatch and the ecosystem at large.

Caveats / Uncertainties

- Climate change impact was simulated only through variations in the biomass at the base of the food web (phytoplankton). Other changes, such as increasing water temperature or decreasing oxygen, and their direct impacts on species were not considered.
- Current climate simulations only accounted for a limited set of uncertainties. Future analyses should consider uncertainty in biogeochemical and trophic processes to fully capture the likely impacts of climate change on edible bycatch.
- Fishing scenarios tested were designed to identify trends in trophic structure changes. Next steps should consider fishing scenarios co-developed in partnership with stakeholders as well as additional considerations around catch efficiency and bycatch mitigation measures.